



CN PROJECT REPORT

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SEMESTER:

3(A)

DEPARTMENT:

COMPUTER SCIENCE

SUBJECT:

COMPUTER NETWORK

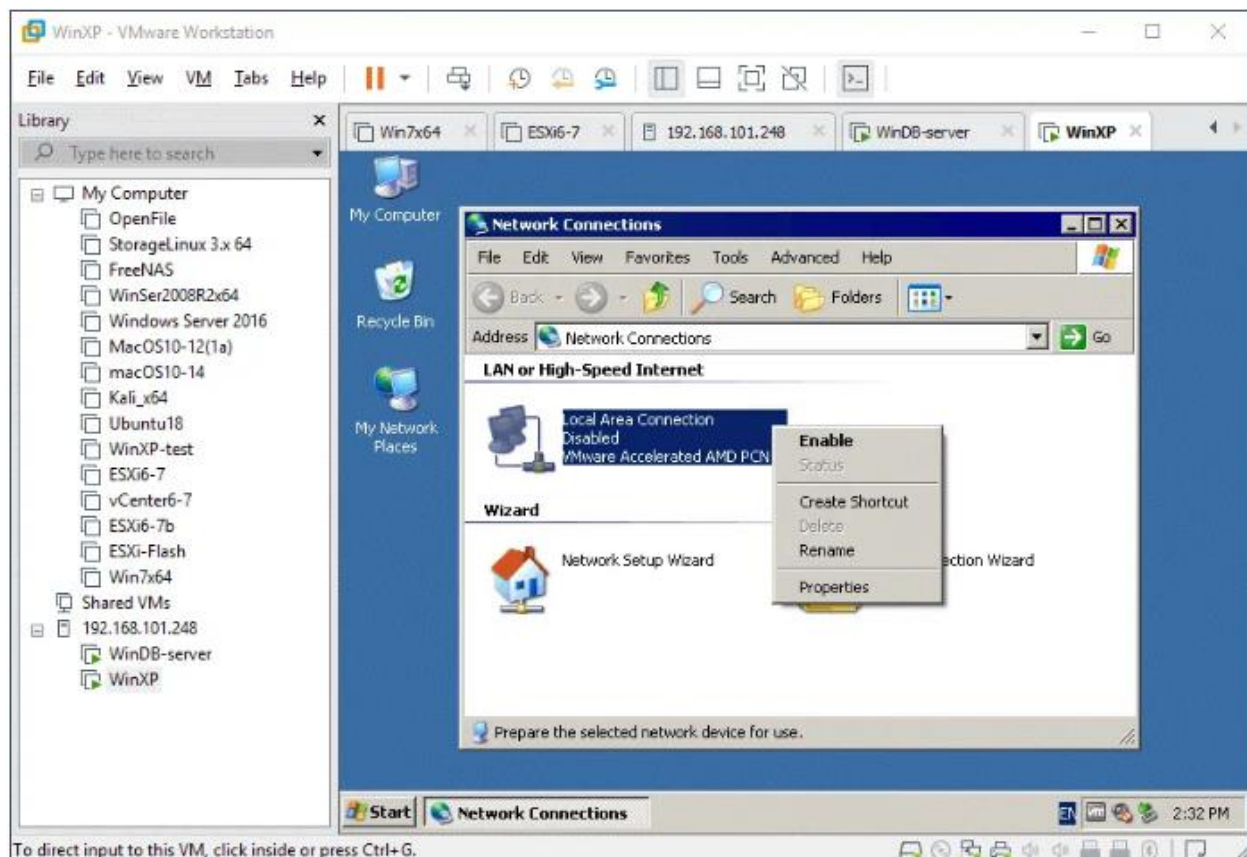
INSTRUCTOR:

SIR. MAJID SHAFIQUE

Project Report: Setting Up VMware and Server Connection

1. Introduction:

The goal of this project was to demonstrate the process of setting up a virtual machine (VM) using VMware, installing a server OS version from 2015 or later, and connecting the virtual machine to a physical machine (PV). The use of VMware for virtualization enables testing network configurations in isolated environments, offering the flexibility to simulate various network scenarios without requiring additional physical machines. This setup can be used for network simulations, software testing, or managing network services.



Objective:

- Install VMware Workstation/ESXi to create virtual machines on a host machine.
- Set up and configure a server operating system (Windows Server 2016, 2019) or Linux-based OS on the virtual machine.
- Establish a network connection between the virtual machine and a physical machine, testing network configurations and connectivity.

2. System Requirements:

The successful setup of VMware and the virtual machine requires specific hardware and software components. Below are the essential requirements to ensure smooth installation and operation.

Hardware Requirements:

- **CPU:** A 64-bit processor with virtualization support (Intel VT-x or AMD-V). An Intel Core i5 or higher is recommended for running multiple VMs.
- **RAM:** At least 8 GB of RAM (16 GB or more is recommended for better performance, especially when running multiple virtual machines simultaneously).
- **Storage:** At least 100 GB of free disk space on the host machine to store the virtual machine and server OS files.
- **Network Adapter:** A functional network interface card (NIC) on the physical machine, along with virtual NICs for the VM (configured within VMware).

Software Requirements:

- **VMware Workstation:** VMware Workstation Pro or Player (version 15 or higher), or VMware ESXi for advanced server-based virtualization.
- **Server OS:** A server operating system version from 2015 or later (e.g., Windows Server 2016, 2019, or Ubuntu Server 20.04 LTS).
- **Network Tools:** Software tools for testing network connectivity, such as ping, traceroute, and network file sharing protocols (e.g., SMB for Windows or NFS for Linux).

3. Installation of VMware:



VMware Workstation allows the creation and management of multiple virtual machines on a single host. It supports various operating systems, and the installation process is straightforward.

Steps:

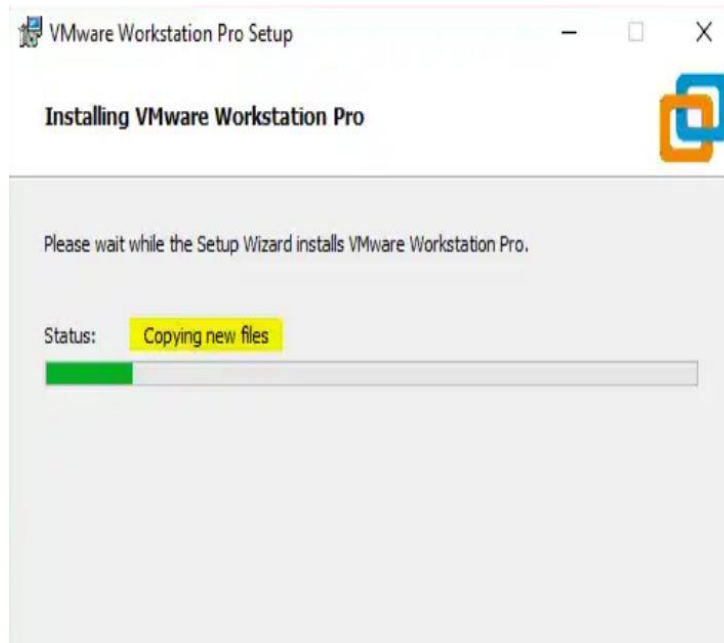
1. Download VMware Workstation:

- Visit the VMware official website and download the VMware Workstation version compatible with your host OS (Windows or Linux).
- Ensure your system meets the minimum hardware requirements for running VMware.



2. Install VMware Workstation:

- Run the installation package and follow the instructions provided by the setup wizard.
- During installation, VMware will configure its network adapters, and you may be asked to install additional drivers.
- Upon successful installation, VMware Workstation will be ready for use.

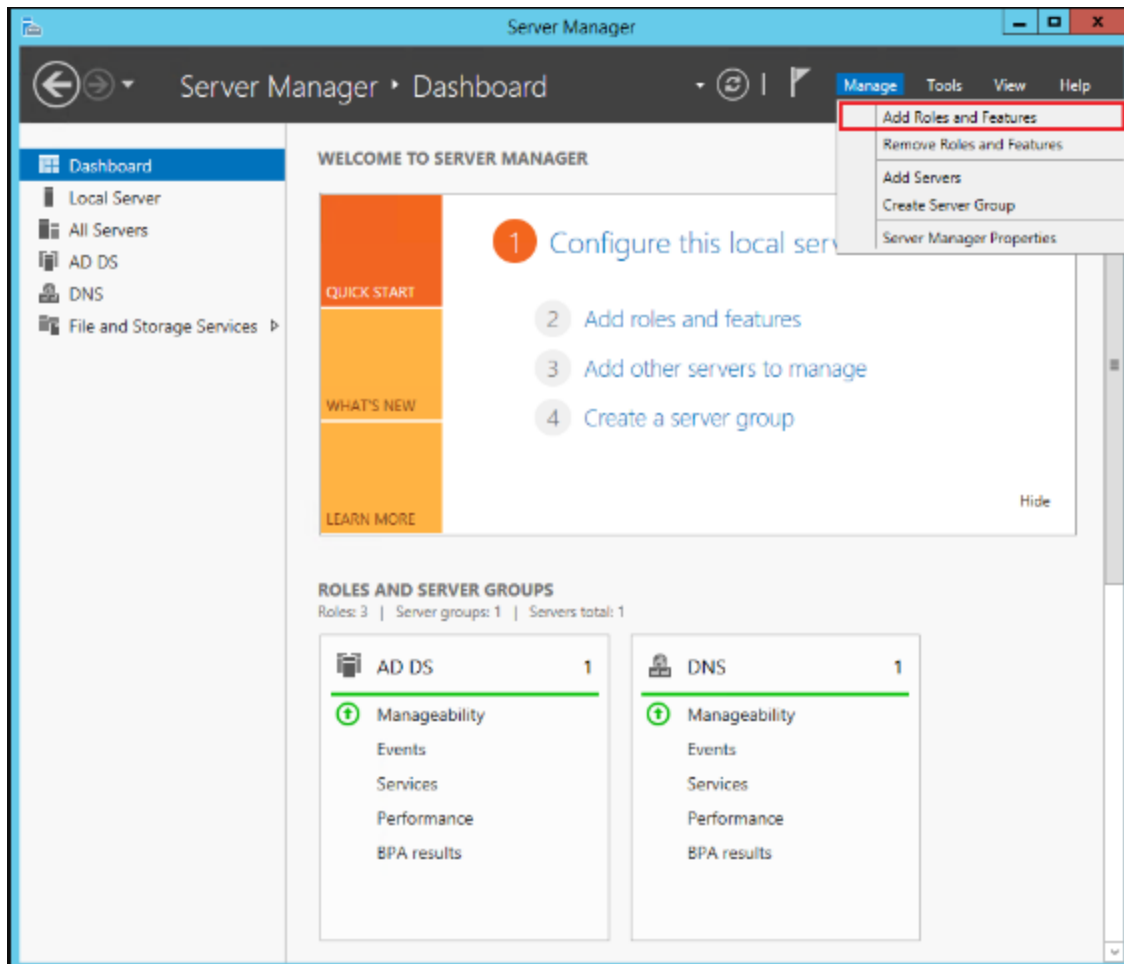


3. Post-Installation Configuration:

- Open VMware Workstation and configure default settings for virtual machines, such as memory size, CPU allocation, and disk space.
- Ensure VMware is properly integrated with the host system's networking settings, particularly if you plan to connect virtual machines to a physical network.

4. Setting Up the Server (2015 or Later):

For this project, we opted to install **Windows Server 2019**, a modern server operating system, on the virtual machine.



Steps:

4. Create a New Virtual Machine:

- Launch VMware Workstation and choose the "Create a New Virtual Machine" option.
- Select the "Typical" installation option for a simplified setup.
- Point to the Windows Server 2019 installation ISO file (or use a physical DVD if available).
- VMware will automatically detect the OS version and configure the virtual machine's hardware settings based on it.

2. Configure Virtual Machine Settings:

- **CPU and RAM:** Allocate at least 2 CPU cores and 4 GB of RAM to the virtual machine.
- **Disk Space:** Allocate at least 50 GB of storage for the virtual machine. You can increase this later based on the server's needs.

- **Network Adapter:** Set the network adapter to "Bridged" for the VM to obtain an IP address from the same network as the host machine, allowing seamless communication with physical machines.

3. Install the Operating System:

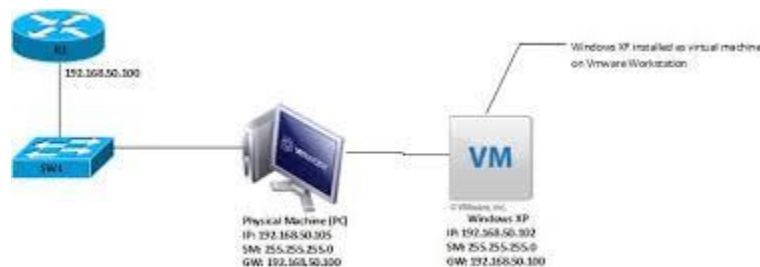
- Once the VM is created, power it on, and the Windows Server 2019 installation process will begin.
- Follow the prompts to set the system language, keyboard layout, and other basic configurations.
- Choose the "Custom Installation" option and select the virtual hard drive where Windows Server will be installed.
- After installation, the system will reboot and prompt for basic configurations like user credentials, time zone, and network settings.

4. Post-Installation Configuration:

- Set a static IP address for the server (e.g., 192.168.1.100).
- Enable Windows Firewall and configure inbound rules to allow remote desktop and other required services.
- Optionally, install additional features like DNS, DHCP, or File Server roles based on your requirements.

5. Connecting the VM to a Physical Machine:

To establish a network connection between the virtual machine and a physical machine, VMware allows you to configure network adapters in various modes such as Bridged, NAT, and Host-Only.



Steps:

1. Configure Network Adapters in VMware:

- Open the VM settings in VMware Workstation and navigate to the "Network Adapter" section.

- Choose **Bridged Networking** to make the VM part of the same network as the physical machine. This allows both machines to communicate using their own IP addresses in the same subnet.
- Alternatively, use **NAT** if the physical machine's network configuration needs to be shared with the VM but without exposing the VM directly to the external network.

2. Verify Network Configuration:

- Set a static IP on the physical machine and ensure it's in the same network range as the VM (e.g., both machines have IP addresses in the 192.168.1.x range).
- Make sure that both devices have the correct subnet mask (typically 255.255.255.0) and DNS settings.

3. Test Connectivity:

- From the physical machine, open Command Prompt and type ping [VM IP address] to check if the VM responds.
- From the VM, use ping [Physical Machine IP address] to ensure the connection is bidirectional.
- If either ping test fails, verify network settings on both machines, firewall rules, and the correct configuration of network adapters in VMware.

6. Network Configuration:

Proper network configuration ensures that both the virtual machine and physical machine can communicate effectively. This section covers assigning IP addresses, setting subnet masks, and configuring firewalls for secure communication.

Steps:

1. Static IP Configuration:

- Configure both the VM and physical machine to use static IP addresses within the same subnet. For example:
 - Physical Machine: 192.168.1.10
 - Virtual Machine: 192.168.1.20
- Set the subnet mask to 255.255.255.0, and use a DNS server (e.g., Google DNS 8.8.8.8) or the local network's DNS server.

2. Firewall Configuration:

- On both machines, ensure that the firewall allows traffic from the other machine. For instance, enable **File and Printer Sharing** and allow **Remote Desktop** through the firewall.

- On Windows, use the “Allow an app through Windows Firewall” option to enable network services.

3. Test Network Connectivity:

- Perform a **ping** test to verify network connectivity.
- Test file sharing or remote desktop functionality to ensure full network integration.

7. Testing and Troubleshooting:

Testing network connectivity and troubleshooting common issues are essential to ensure everything works smoothly.

Tests:

- **Ping Test:** Ping the VM from the physical machine and vice versa. If the ping test is successful, the basic network setup is correct.
- **File Sharing:** Set up shared folders on both the VM and the physical machine and attempt file transfers to test bidirectional communication.
- **Remote Desktop:** If the Windows Server VM supports Remote Desktop, attempt to access the VM from the physical machine using the Remote Desktop Protocol (RDP).

Troubleshooting:

- **Ping Test Fails:** If the ping fails, check if the network adapter in VMware is configured correctly. Ensure the VM and physical machine are on the same network subnet.
- **Firewall Issues:** Check firewall rules on both machines. Temporarily disable the firewall to see if it's blocking communication.
- **Incorrect IP Addressing:** Verify that IP addresses on both the VM and physical machine are correctly assigned and belong to the same subnet.

8. Conclusion:

This project successfully demonstrated the process of setting up VMware, configuring a server OS on a virtual machine, and connecting it to a physical machine via a bridged network. By using VMware Workstation, we created a flexible and isolated environment for testing network configurations. The connection between the virtual machine and physical machine allowed for practical testing of network connectivity and communication, providing insights into virtualization and networking concepts.

9. References:

- VMware Workstation Documentation: VMware Official Docs
- Windows Server 2019 Installation Guide: [Microsoft Docs](#)

- Networking Basics: Cisco Networking

❖ Contributions by Sania Arshad (098)

Sania focused on identifying and documenting the hardware and software requirements essential for the project. Her contributions included:

1. **System Requirements:**
 - Documenting the hardware requirements such as CPU, RAM, storage, and network interface card (NIC) for virtualization.
 - Listing software requirements, including VMware Workstation, supported server OS versions (e.g., Windows Server 2016, 2019), and network testing tools (ping, traceroute).
 2. **Research and Recommendations:**
 - Ensuring the selected tools and OS versions were compatible with the project's objectives.
 - Providing insights into advanced virtualization software like VMware ESXi.
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❖ Contributions by Eman Waheed (027)

Eman Waheed handled the installation and configuration of VMware Workstation. Her contributions included:

1. **Installation of VMware Workstation:**
 - Downloading VMware Workstation Pro from the official website and ensuring hardware compatibility.
 - Installing the software by following the setup wizard and configuring necessary drivers.
 2. **Post-Installation Configuration:**
 - Setting default configurations for virtual machines, such as memory size, CPU cores, and disk space.
 - Integrating VMware's network adapters with the host machine's network settings.
 3. **Documentation of Installation Steps:**
 - Providing detailed steps for downloading, installing, and configuring VMware for the project.
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❖ Contributions by Malaika Arooj (057)

Malaika focused on creating and configuring the virtual machine and installing the server OS. Her contributions included:

1. **Setting Up the Virtual Machine:**

- Creating a new virtual machine in VMware and selecting a "Typical" setup.
 - Allocating appropriate CPU, RAM, and disk space based on project requirements.
 - 2. **Installation of Server OS:**
 - Installing Windows Server 2019 using an ISO file and following the guided setup process.
 - Configuring basic settings like static IP addresses, time zones, and network configurations.
 - 3. **Post-Installation Server Setup:**
 - Enabling essential server roles such as file sharing and remote desktop services.
 - Configuring firewall rules for secure communication.
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❖ **Contributions by Maryam (061)**

Maryam worked on connecting the virtual machine to the physical machine and verifying the network setup. Her contributions included:

1. **Network Configuration:**
 - Setting the VMware network adapter to "Bridged" mode to enable communication between the VM and the physical machine.
 - Assigning static IP addresses to both the VM and physical machine, ensuring they were in the same subnet.
 2. **Testing and Troubleshooting:**
 - Conducting ping tests to verify bidirectional connectivity between the VM and physical machine.
 - Troubleshooting network issues such as firewall blocks and incorrect IP configurations.
 3. **Documentation of Network Setup:**
 - Providing a detailed guide on configuring network adapters, assigning IP addresses, and testing connectivity.
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➤ **Collaborative Contributions (All Members)**

The team collaborated on:

1. **Finalizing Network Configuration:**
 - Assigning IP addresses, configuring firewalls, and verifying network integration.
 - Testing additional features like file sharing and remote desktop functionality.
2. **Testing and Troubleshooting:**
 - Performing connectivity tests, resolving issues, and ensuring seamless communication between the virtual and physical machines.
3. **Report Writing:**

- Combining individual contributions into a comprehensive and cohesive project report.